

PRITAM KUMAR PANDA

[in LinkedIn](#) | [734-637-6338](tel:734-637-6338) | atomodyssey.com | pritam@stanford.edu | [GitHub](#) | [Google Scholar](#)

Skills

- **Languages & ML Frameworks:** Python | PyTorch | JAX / JIT | Warp | TensorFlow | Hugging Face | scikit-learn | MLflow | R | SQL
- **Deep Learning:** Graph Neural Networks (GNNs) | Generative Design | Diffusion Models | BioEmu | Transformers
- **Atomistic & Molecular Modeling:** GROMACS | Amber | OpenMM | TorchSim | JAX-MD | Umbrella Sampling | Steered MD | Free Energy Perturbation (FEP) | Conformer Generation | CHARMM-GUI | VASP | Siesta
- **Drug Discovery & Cheminformatics:** AutoDock4/Vina | AutoDock-GPU | NVIDIA-DiffDock | AlphaFold2/AlphaFold3 | Boltz2 RFdiffusion | RDKit | QSAR Modeling | Schrödinger | ChimeraX | CADD | ADMET Profiling
- **Multomics/NGS:** WES/WGS, Variant Calling | RNA-seq (DESeq2, edgeR) | Single-Cell (Seurat/scanpy) | Spatial Transcriptomics Proteomics/ Metabolomics/Microbiome/Epigenomics | Biomarker Discovery | Data Harmonization & Cross-platform Integration
- **Cloud & HPC Infrastructure:** Nextflow | Snakemake | Docker/Singularity | AWS (EC2/S3/HealthOmics/ CloudWatch) | CUDA/CuPy Multi-GPU Training | HPC/SLURM | Performance Profiling | NVIDIA Nsight | Nebius AI | Data/Tools/Software Benchmarking

Experience

Postdoctoral Scholar

[Stanford University](#)

Palo Alto, CA, USA 01/2025 - Current

- Architected an **AI-driven drug discovery platform** that advanced a DoD/DARPA anesthesia program to **Phase II clinical trials**, integrating multi-parameter optimization screening of 2.5 billion small-molecules with RDKit, Schrödinger and Discovery Studio.
- Engineered a scalable **AlphaFold-enabled structural modeling pipeline** (PyTorch/JAX) to generate 100,000+ neurotransmitter receptor conformations; trained multimodal foundation models for structure-function learning and generative design, leading to identification of candidate targets for slow-type GABAergic therapeutics.
- Implemented **GPU-accelerated atomistic simulation** workflows integrating GROMACS MD, alchemical FEP (mutational FEP/PMX), umbrella sampling (PMF), and residue-level energy decomposition to mechanistically model mutation-driven modulation of chloride ion conductance in neurotransmitter-gated ion channels.
- Coupled simulations with **active learning-based docking triage** (AutoDock-GPU/Vina/Glide/CDOCKER/Boltz2), ADMET & QSAR profiling, reducing experimental candidate testing by 60%.

Data Manager | Bioinformatician

[German Cancer Research Center](#)

Heidelberg, Germany 07/2023 - 12/2024

- Supported high-throughput multi-omics analysis operations (WES, WGS, RNA-seq, scRNA-seq, metabolomics, ATAC-seq), resolving ~20 client-facing technical issues daily within an **Agile/Scrum ticketing framework**, improved pipeline reliability and turnaround time through systematic debugging and workflow optimization.
- Designed and maintained **reproducible bioinformatics workflows** (nf-core/raredisease/sarek/variantbenchmarking/rnaseq/scrnaseq/atacseq) using **Nextflow**, **Snakemake**, and **Bash**, containerized with Docker/Singularity for scalable execution across HPC environments.
- Contributed to development of the HI-TRON data portal, implementing **Vue.js frontend** and **R/Python backend** services to host and manage multi-project omics datasets under FAIR data principles.
- Delivered internal technical lectures and authored **SOPs** for workflow execution, pipeline validation, and client onboarding, standardizing operational best practices across teams.

Affiliated Researcher

[Karolinska Institute](#)

Stockholm, Sweden 03/2021 - 12/2022

- Developed **predictive ML models** for patient stratification using genomic and antibody profiling, contributing to **2 peer-reviewed publications** and securing **1 patent** ([US20250339544A1](#)).
- Engineered **scalable Immunoinformatics pipelines** processing 2TB+ multimodal clinical datasets, adopted to support **3 ongoing clinical trials**, accelerating translational research workflows.

PhD Researcher

[Uppsala University](#)

Uppsala, Sweden 11/2018 - 03/2023

- Originated **GENOME2QUNOME**, the first quantum-to-biological modeling framework (physics+bioinformatics+ML) bridging DFT-based 2D material simulations with atomistic modeling enabling predictive design for nanopore sensing and bio-inspired materials.
- Conducted **first-principles DFT and ML-augmented** electronic structure simulations (VASP, SIESTA/TranSIESTA) to tune bandgaps, characterize charge transport, and guide material selection in novel 2D and ZnO tetrapod nanostructures; conducted exploratory docking of **ZnO tetrapods** against viral targets (Herpes virus) to identify potential antiviral interactions for therapeutic exploration.
- Led **industry-collaborative ab initio studies** on thermal and electronic transport in cellulose-based materials for transformer applications (Hitachi ABB), developing mechanistic insights that informed energy-efficient bio-inspired materials.

NGS Data Analyst

[Freiburg Medical Center](#)

Freiburg, Germany 09/2017 - 10/2018

- Engineered **high-throughput WES/WGS variant calling pipelines**, processing 200+ clinical samples/month with HPC parallelization and dependency-aware job scheduling, reducing manual runtime by **70%** and accelerating analysis for precision oncology projects.
- **Enhanced STAR aligner performance and HPC memory management**, improving alignment speed **2.5x** while reducing compute overhead, enabling faster and more cost-effective genomics processing at scale.
- **Co-authored [Nature Medicine publication](#)** with the European MDS Working Group on clonal hematopoiesis trajectories, demonstrating translational impact of computational genomics on **clinical decision-making**.

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Education

PhD [Uppsala University](#) Uppsala, Sweden **11/2018 - 03/2023**

- Specialized in Quantum Physics, Computational Biophysics, DFT, Condensed Matter Physics

Master of Technology [D Y Patil](#) Mumbai, India **07/2014 - 07/2016**
Master of Science [Utkal University](#) Bhubaneswar, India **07/2012 - 06/2014**

- Major in Bioinformatics, Computational Biology

Publications & Conferences

- 50+ peer-reviewed publications (5,500+ citations, H-index: 46, i10-index: 102)**
- Selected publications:** *Science Advances* (2020): Structure-based drug design and immunoinformatics for SARS-CoV-2 | *Nature Medicine* (2021): Clonal hematopoiesis trajectories in SAMD9/SAMD9L syndromes | *Nature Leukemia* (2019): Targeting leukemia-initiating cells in JMML | *Materials Today* (2023): Nanocarrier vaccine therapeutics | *Experimental Dermatology* (2024): Computational biology-guided clinical dermatology study | *Journal of Molecular Biology* (2021): Dynamics of Amyloid- β
- Conferences:** **Invited speaker / participant** at Helmholtz Health Summit: *The Future of Medicine* (Berlin) | European Working Group of MDS (EWOG-MDS) | Blood (ASH) Annual Meeting | HITRON Symposium (Mainz) | Curious Future Insight Conference (Berlin)

Open-Source Contributions

- PandaDock:** Next-generation molecular docking platform with SE(3)-equivariant Graph Neural Network scoring, achieving $R > 0.8$ correlation with experimental binding affinities. (03/2025)
- BioClaw:** BioClaw combines a Claude-powered ReAct agent with a curated toolkit of 17 bioinformatics tools (NGS + Drug Discovery), accessible through a CLI, REST API, or Next.js web interface. (02/2026)
- nf-core/moleculardocking & nf-core/moleculardynamics:** Contributed production-grade Nextflow pipelines for virtual screening and molecular dynamics simulations, enabling reproducible HPC execution and automated analysis. (01/2026)
- ADMET Prediction System (GNN-RAG):** Graph neural network with retrieval-augmented generation for ADMET and drug-likeness prediction. (11/2025)
- PandaMap:** Interactive protein-ligand visualization, integrating 2D ligand representations with 3D docking outputs. (04/2025)
- MDInsight:** Machine learning toolkit for molecular dynamics trajectory analysis and conformational sampling. (01/2026)
- Oncology Drug Response (DL-RAG):** Deep learning + RAG system for predicting patient drug response from multi-omics datasets with end-to-end evaluation pipeline. (11/2025)
- Swift iOS apps: ChemStudio and MDStep** for enabling ADMET profiling and for molecular dynamics users (05/2025)
- Additional Open-Source Tools: 10+ repositories including **Drug Repurposing (R-GCN)**, **Literature Intelligence (BioBERT)**, and nf-core workflow enhancements; actively maintained and publicly available on [GitHub](#). (2025-2026)

Mentorship

- Scientific Advisor and mentor** for [Revilico, Inc](#) guiding research strategy and technical roadmap for a computational drug design platform, advising teams on translating open-source innovations into **production-grade, scalable discovery pipelines**.
- Co-founded** and initiated early-stage startup initiatives, demonstrating ownership, execution discipline, and cross-functional leadership in research-driven innovation. (12/2021-07/2023)
- Led and **coordinated a global team of 160 researchers across 10 countries** to design and deliver an **international longevity hackathon**, overseeing program structure, mentorship coordination, and execution. (06/2021-09/2022)
- Instructor for **Master's-level coursework** at KIIT University (India), designing curricula, lesson plans, and interactive sessions to strengthen student proficiency in computational biology and molecular modeling. (08/2022-01/2023)
- Bioinformatics Tutor:** Drug Designing | NGS/Multiomics | Bioinformatics Workflows | Quantum Physics | Career Advice (2019-present)

Recognitions

- Elected member, **Sigma Xi – The Scientific Research Honor Society**, recognizing significant scientific contributions. (09/2024)
- Appointed **Nextflow Ambassador** by **Seqera and the nf-core community** for leadership in workflow standardization and community engagement. (07/2024)
- Scientific reviewer for 30+ proposals for the Biotechnology Industry Research Assistance Council (BIRAC), contributing to national funding decisions. (06/2023)
- Recipient of **competitive research grants** including, Colgate & Palmolive (CADD research) | IMM Strategic Interdisciplinary Collaboration Grant (Karolinska Institute) | AIMday Materials 2020 Innovation Grant (ABB Power Grids / Hitachi, Uppsala University) (2021-2023)
- Built a scientific outreach platform with **17,000+ engaged LinkedIn followers**, sharing insights on bioinformatics, molecular modeling, drug discovery, and computational workflows. (2017-2026)
- Invited expert guest for one-on-one discussions on leading Indian [TV news](#) channels; featured in *The Global Indian* series [“Indians in Europe”](#) for interdisciplinary research journey. (06/2023)

Certifications

- NVIDIA Foundation Models in Digital Biology | Google MLOps for Generative AI/Vertex AI: Model Evaluation | AWS Cloud Practitioner